TECHNICAL & LEGAL COMPLIANCE WHITEPAPER

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FGF GUILD MANAGEMENT

Technical Architecture, Security, Privacy by Design & Intellectual Property Compliance Whitepaper

Purpose of this Document

Prepared for the **Game Publisher, Developer, Legal Counsel and Community Operations teams**. It describes an independent, free, out-of-game companion tool that does **not interact with the game client or its network traffic**, uses **no copyrighted game assets**, processes a **limited, published set of personal data**, and isolates every guild with **PostgreSQL 17 Row-Level Security**.

0	0	0	100%
COPYRIGHTED GAME ASSETS	GAME MEMORY / CLIENT HOOKS	PAYMENTS, COOKIES OR ANALYTICS	PUBLIC TABLES UNDER RLS + FORCE RLS

TARGET AUDIENCE
Game Publisher & Developer Compliance / Security Teams

DOCUMENT VERSION
3.0 (supersedes the August 2026 edition)

FRAMEWORKS
GDPR (EU 2016/679), OWASP Top 10, Zero-Trust Access

REVISED
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What changed since the August 2026 edition

That edition claimed the platform held zero personal data, that IP addresses were never persisted, that deleting a profile cascaded through every record, that payments went through Stripe, that the app was an installable PWA with an offline cache, and that screenshots and recordings never touched storage. None of that is true today, and some of it was never true. The app is free with no payment surface (ADR-015); the PWA and Web Push were removed (ADR-016). Every statement below was checked against the repository or the production database on the revision date; the data inventory follows ADR-018 and the published Privacy Policy, which prevail in case of disagreement.

1. Executive Summary & Scope

1.1 Purpose of the Platform

The **FGF Guild Management Platform** is an out-of-game web application that helps guild leaders, officers and players of *Foundation Galactic Frontier* organise rosters, schedule guild events, track attendance and scores, and post announcements to their own Discord servers. It is free for every guild, with no paid tier.

The platform runs **entirely outside** the game client and the game servers. It works in the same capacity as a shared spreadsheet or a community Discord bot.

1.2 Foundational Principles

Game Integrity & Non-Interference

- ✕ **No client hooks**: no injection into game memory or processes.
- ✕ **No network tampering**: no packet interception.
- ✕ **No automation**: no bots, macros or gameplay actions.
- ✕ **No private APIs**: inputs are what users type or upload.

Data & IP Safeguards

- ✓ **Data minimisation**: pseudonyms, player IDs, Discord IDs if linked, 24-hour sign-in IPs.
- ✓ **No game assets**: open-source fonts and icons, self-hosted.
- ✓ **OCR**: screenshots not stored; recordings staged privately, then deleted.
- ✓ **PostgreSQL 17 RLS**: database-enforced tenant isolation.

2. Personal Data & Privacy by Design (GDPR)

The operator is established in France, so the GDPR applies whether or not the service is paid. The full notice is the Privacy Policy (`privacy.html`), whose inventory is exhaustive.

2.1 Data Inventory

DATA CATEGORY	PROCESSED?	DETAIL
In-game identity	YES	Pseudonym, in-game player ID (UID), pseudonym-change history (12 months). Online identifiers under GDPR Art. 4(1).
Game activity	YES	Attendance, scores, power and ratings, absences, squads, sanctions and bans.
Account data	YES	Account identifier, role, guild and server; password stored as a bcrypt hash. A legacy reversibly encrypted copy remains only for accounts not signed in since 2026-10-03, deleted at their next sign-in.
Discord identifiers	IF LINKED	Discord user ID and username; OAuth scope `identify` only (no email, no server list). Call-to-arms answers record the Discord user ID for 7 days.
IP addresses	24 HOURS	Sign-in IP stored in `public.login_throttle` for rate limiting, purged hourly by the `login-throttle-retention` job (rows older than 24 hours). No analytics, profiling or geolocation.
Free text	YES	Support tickets, wiki pages and comments.
Diagnostics	ON ERROR	Error reports to Sentry (EU): error, code path, active guild; credentials stripped.
Player email addresses	NO	None collected. Each account's sign-in identity uses a synthetic address derived one-way from the account identifier.
Names, address, phone, birth date, ID documents	NO	Never requested.
Payment data	NO	The app is free: no checkout, subscription or donation path (ADR-015).
Game credentials	NO	Game, Apple or Google credentials are never requested.
Location / special categories	NO	No geolocation; no Art. 9 or Art. 10 data.

Retention

Enforced by scheduled database jobs, mainly `gm_apply_data_retention()` daily at 04:30 UTC: sign-in IPs 24 hours; codes, tickets and cached OCR text until expiry; call-to-arms answers 7 days; unactioned guild requests 14 days, their Discord handle erased 90 days after closure; audit log and pseudonym history 12 months; rosters and statistics while the guild uses the service.

Erasure

On request, through `gm_erase_player` and `gm_erase_account`. A player is **anonymised**: the pseudonym is replaced by a token from a salt that is never stored, UID and timezone cleared, absences and history deleted; the roster row survives unattributable so guild totals stay valid. Ban records are kept in full, as disclosed (Art. 17(3)).

No Tracking

No cookies, analytics, advertising SDKs or pixels. Browser storage holds only session and display state, strictly necessary, so no consent banner is required. The CSP `connect-src` limits the browser to the Supabase project and Sentry ingest.

Processors

Supabase (database, auth, storage, functions; EU database), Vercel (hosting), Google Gemini (OCR, wiki summaries), Discord (sign-in, notifications), Sentry (EU, diagnostics), Render (verification worker). Transfers outside the EU under standard contractual clauses or the EU–US Data Privacy Framework.

3. Intellectual Property, Asset & Trademark Safeguards

3.1 No Game Assets

- ✕ **No game textures or art**: no ship, hero or item graphics from the game.
- ✕ **No 3D models**, **no audio**, **no game code or shaders**.

3.2 Interface Provenance

COMPONENT	SOURCE	LICENSE
Design tokens	Original token set (`tokens.css` , `theme.css`) on Material Design 3 conventions	Original work
Typography	Inter, Plus Jakarta Sans, JetBrains Mono, self-hosted	SIL Open Font License 1.1
Icons	Phosphor Icons (`@phosphor-icons/web`), subset font served locally	MIT
Charts	Hand-written rendering (`PortalChart.ts`)	Original work
Game assets	None	—

The CSP allows scripts, styles and fonts from the app's own origin only: no external font or script CDN, and no `'unsafe-inline'` (no inline script or style attribute since 2026-10-04).

3.3 OCR Processing

1. **Screenshots** travel in the request body over TLS to the `ocr-guild-members` Edge Function, are read by Google Gemini for the duration of the request, and are **not stored**. The transcribed text and a SHA-256 fingerprint of the image are cached in `public.ocr_cache` until expiry.
2. **Screen recordings** (MP4, MOV or WebM; at most 64 MiB and 300 s) are uploaded by the browser to the private Storage bucket `ocr-videos` , where RLS lets only that guild's administrators upload, read or delete. The function reads the video with the caller's own token, sends it to the Gemini File API, then deletes the Gemini file and the stored object whatever the outcome; recordings left over from an interrupted scan are removed at the guild's next scan after one hour. Google deletes the file after processing or within 48 hours.
3. **Review before write**: the officer sees a diff against the roster and confirms; a roster import is all-or-nothing.

Non-Affiliation

The README and the Terms state that this is an unofficial, community-developed project, not affiliated with or endorsed by the publisher of *Foundation Galactic Frontier*, and that all game names and trademarks belong to their owners.

4. Technical Stack



Client

- Strict TypeScript, Vite
- Views loaded on first use; admin code in its own chunk
- One global only: the frozen e2e entry point
- No PWA; a self-unregistering tombstone service worker

Edge Functions

- Deno on Supabase
- 14 in the repository, 14 deployed (ADR-017)
- Deployed from CI after every check passes
- Zod-validated inputs

Database

- PostgreSQL 17 on Supabase
- RLS + FORCE RLS on every table
- Migration history replayed in CI
- Audit log kept 12 months

4.1 Logging & Audit

- **Structured logs**: `EdgeLogger` emits JSON with correlation IDs and redacts credentials and tokens; browser errors go through a rate-limited RPC and Sentry with PII scrubbing.
- **Audit trail**: administrative actions write to `public.system_audit_logs`, whose insert is reserved to the service role.

5. Security Model

5.1 Four Roles

Roles are stored in `public.accounts.role`, limited by a CHECK constraint to four values, and resolved from `auth.uid()` on every request — never from client-supplied data.

ROLE	SCOPE	ENFORCEMENT
`super_admin`	All guilds; account provisioning, audit log, ticket console.	`accounts` row resolved from the JWT.
`server_admin`	Guilds on their own game server.	RLS helpers compare `guilds.server_number`.
`guild_admin`	Their own guild only.	`gm_can_read_guild_data`, `check_user_guild_write_access`.
`member`	No direct table access; the Player Portal only.	Every RLS read helper returns false; requests go through the `member-portal` Edge Function.

5.2 Row-Level Security

Every public table has RLS **and** FORCE RLS. Policies call `SECURITY DEFINER` helpers with `SET search_path = ''`, which also refuse a session whose epoch was revoked (password reset, role change, suspension). Tables holding credential material are granted per column.

> **No subscription gate**
> Write access is decided by `check_user_guild_write_access(guild)` alone. The former `is_subscription_active()` was an inert stub (a literal `SELECT true`) still called by 45 write policies; it gated nothing and was removed and dropped on 2026-10-04. The app is free.

5.3 Player Portal Isolation

A player signs in with Discord and joins a guild with a join code and their UID, approved by an officer. Direct table queries from a member return nothing. The portal calls `member-portal`, which requires a platform-verified JWT, resolves the account from it (active status and current session epoch required) and derives the player's UID server-side; client-supplied identities are ignored.

5.4 Edge Function Protections

- **Authentication**: `verify_jwt` is set per function in `supabase/config.toml`. Seven functions take no Supabase JWT because no session exists yet or the caller is not a user (sign-in, registration, guild requests, the cron-triggered reminder dispatcher, the Discord interactions endpoint verified by Ed25519 signature, the verification worker authenticated by a shared secret); the other seven require one.
- **Input validation**: Zod schemas; scores bounded to 0–9,999,999,999.
- **Rate limits** fail closed (sign-in, portal, OCR video budget).
- **Webhooks**: Discord webhook URLs are stored server-side in `guild_config`, never supplied by the client at send time; only Discord hosts over HTTPS are accepted.
- **Secrets** live in Supabase Vault or function environment variables, never in the repository.

6. Quality Assurance

Every push runs the CI pipeline; deployment jobs run only after all checks pass.

```
1. npm run type-check      zero TypeScript errors
2. npm test                Vitest suite, with a coverage floor
3. npm run build           production bundle, with an entry-size budget
4. npx playwright test     end-to-end tests against the built bundle in dist/,
                           incl. axe-core accessibility checks, CSP and asset errors
5. deno check / deno test  Edge Functions against an in-memory Supabase
6. migration replay        PostgreSQL 17 from scratch, schema fingerprint vs production,
                           SQL tests of RLS, grants and RPCs as each role
```

On 2026-10-04 the suite stood at 2,508 Vitest tests and 61 Playwright tests (CHANGELOG, phase M5). The number grows with every change: read the CI run rather than this line.

7. Publisher Compliance Matrix

AREA	REQUIREMENT	IMPLEMENTATION	STATUS
Personal data	Lawful, minimal processing (GDPR).	Published inventory, retention jobs and erasure functions (ADR-018). No player emails, payment or location data.	DOCUMENTED
Game modification	No memory injection, packet tampering or hooks.	External web application; no interaction with the game client.	YES
Automation	No automated gameplay.	Manual management and voluntary score entry; the Discord bot acts in Discord only.	YES
Copyrighted assets	No proprietary textures, models, audio or binaries.	OFL fonts, MIT icons, original tokens and charts.	YES
Screenshots & recordings	Not archived or monetised.	Screenshots not stored; recordings staged privately and deleted after transcription.	YES
Access security	Tenant isolation.	RLS + FORCE RLS on every table; four server-resolved roles; session revocation.	YES
Network safety	Safe webhook relaying.	Server-side webhook URLs, Discord hosts only.	YES
Game economy	No currency manipulation or trading.	No virtual goods, trading or game-server writes; no payments of any kind.	YES
Legal status	Clear non-affiliation.	Statement in the README and the Terms.	YES

8. Attestation & Contact

Developer Attestation

The **FGF Guild Management Platform** is an independent community tool. It supports guild coordination without touching the game client or servers, uses no publisher assets, and processes the limited personal data described in Section 2 under a published Privacy Policy.

Contact

Controller and maintainer: André Vieira, **fgfwiki@gmail.com** (the single contact address in the Privacy Policy). Security and compliance concerns sent there are handled with priority.